ModelIt!

Teacher's Guide

Why Is Clean Water Becoming Scarce?

Modeling Global Freshwater Systems, Aquifer Depletion, and Water Stress

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-ESS2-5, HS-ESS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-5, HS-ESS3-1

Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of aquifer depletion dynamics and design water management solutions informed by computational model data on recharge-withdrawal balances.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Surface Processes / ESS3.A Natural Resources** — Students model how water moves through the hydrological cycle, accumulates in aquifer systems over geological time, and is depleted by human extraction, connecting water science to resource management.
Crosscutting Concept	**Cause and Effect / Systems and System Models** — Students trace causal chains from population growth and agricultural demand through aquifer depletion to food system collapse, modeling freshwater as a complex system with stocks, flows, and feedback loops.

CAST Testing Alignment

MULTIPLE CHOICE:

The Ogallala Aquifer has lost 30% of its stored water in 60 years. Natural recharge replaces less than 1% of annual withdrawals. At current extraction rates, major portions will be depleted within 25-50 years. Based on your model, which statement BEST describes the situation?

CONSTRUCTED RESPONSE:

Using your model, explain why a region that has abundant rainfall can still face a water crisis. Reference groundwater extraction, recharge rates, agricultural demand, and the difference between renewable and fossil water in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water is the most essential and most overlooked resource crisis of the 21st century. While climate change dominates headlines, the quiet depletion of underground aquifers that took thousands to millions of years to fill threatens the food supply of billions. Understanding global water scarcity requires modeling the interaction between the water cycle, human demand, agricultural systems, and the geological constraints on groundwater replenishment.

Earth's Water Budget

Of Earth's total water, 97.5% is saltwater. Of the remaining 2.5% freshwater, 69% is locked in ice caps and glaciers, 30% is groundwater, and only 1% (0.025% of total water) is accessible surface freshwater in rivers, lakes, and soil moisture. This tiny fraction supports all terrestrial life, agriculture, and human civilization. The water cycle continuously recycles this water through evaporation, precipitation, and runoff — but the rate of cycling is fixed by solar energy and cannot be accelerated by human technology.

Groundwater: The Hidden Resource

Aquifers — underground layers of permeable rock saturated with water — store approximately 10.6 million cubic kilometers of freshwater, about 100 times more than all surface freshwater combined. Some aquifers (like those in limestone karst regions) recharge relatively quickly from rainfall. But many major agricultural aquifers are 'fossil' water — accumulated during wetter climate periods thousands to millions of years ago, with negligible current recharge. The Ogallala Aquifer beneath the U.S. Great Plains, the North China Plain aquifer, and the Upper Ganges aquifer all fall into this category. Pumping fossil water is mining a non-renewable resource.

The Agriculture Water Crisis

Agriculture consumes 70% of all freshwater withdrawals globally — and much of this is profoundly inefficient. Flood irrigation, still used on the majority of irrigated cropland, loses up to 60% of applied water to evaporation and runoff. Shifting to drip irrigation can reduce water use by 30-70% but requires significant capital investment. The global trend toward more water-intensive diets (more meat, more dairy) is increasing per-capita water demand even as population grows. Producing one kilogram of beef requires approximately 15,000 liters of water, compared to 1,800 liters for a kilogram of wheat.

Cascading Consequences of Depletion

When aquifers are depleted, the consequences cascade through multiple systems. First, food production collapses in regions that depend on groundwater irrigation — the Ogallala Aquifer supports $20 billion in annual agricultural production. Second, land subsidence occurs as emptied aquifer spaces compact, permanently reducing storage capacity — Mexico City has sunk over 10 meters in some areas. Third, water scarcity drives migration, creating 'water refugees' who concentrate in cities that face their own water stress. Fourth, competition for remaining water resources creates political instability — the World Bank has warned that water scarcity will be a primary driver of conflict by 2050.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the freshwater system: Population and Agricultural Demand, Groundwater Extraction Rate, Aquifer Water Level, Natural Recharge Rate, Surface Water Availability, and Regional Food Production Capacity — distinguishing human-controlled withdrawal decisions from natural system responses.

E **Establish Relationships**

Students determine that Population Demand drives Groundwater Extraction Rate, which depletes Aquifer Water Level faster than Natural Recharge can replenish it. As Aquifer Water Level drops, extraction costs increase and food production capacity declines. Surface Water Availability is insufficient to replace groundwater, creating a structural deficit.

V **Visualize & Model**

Students build a computational model showing the stock-and-flow dynamics of aquifer systems — the slow trickle of recharge versus the high-volume pump of extraction — visualizing the depletion timeline.

E **Evaluate Outcomes**

Students run sustainable, current overdraft, and climate-change-compounded scenarios to quantify depletion timelines and identify the withdrawal rate that matches recharge for long-term sustainability.

R **Revise & Extend**

Students add Desalination Capacity, Climate Change Impact on Precipitation, or Water Pricing Policy to explore technological, climatic, and economic dimensions of water management.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This picture shows the same farm — ten years apart. The only difference? The water underground ran out. And it's happening across the planet."*

Watch For

Show the GRACE satellite animation of the Ogallala Aquifer shrinking (NASA visualization). Let students watch the red zones expand over 20 years. Ask: What happens when the blue is gone?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling Earth's most overlooked crisis. Not carbon, not temperature — water. And the math is terrifying."*

Watch For

Have students read objectives. Pre-teach 'aquifer' using a sponge demonstration — saturate a sponge (aquifer full), squeeze slowly (extraction), drip water on top (recharge). Show that squeezing is much faster than dripping.

Slide 3: Big Question

3 min

Say: *"70% of Earth is water. But if Earth's water fit in a gallon jug, the freshwater you could actually use would be about one tablespoon. That's it. For 8 billion people."*

Watch For

Physical demonstration: bring a gallon jug of water. Pour out a tablespoon. Hold it up: 'This is all the freshwater we have.' Let it sink in. Quick-write: Who decides who gets that tablespoon?

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a resource that took millions of years to accumulate and decades to drain. A stock-and-flow system where the drain is 100 times bigger than the faucet."*

Watch For

Preview each LEVER step. Emphasize the stock-flow concept: the aquifer is a bank account. Recharge is the paycheck. Extraction is the spending. What happens when you spend 100x your paycheck — every month, for 60 years?

Slide 5: Activity 1: Components

8 min

Say: *"Two human decisions — how much demand we create and how fast we pump. Four system responses. But here's the catch: the biggest system response takes thousands of years to replace."*

Watch For

Guide sorting of external vs. internal components. Emphasize that Groundwater Extraction Rate is the critical human decision. Discuss why Natural Recharge Rate is internal and essentially fixed by geology and climate.

Slide 6: Activity 2: Connections

8 min

Say: *"When extraction exceeds recharge, the aquifer declines. When the aquifer declines, wells must go deeper, costs rise, and eventually — the water simply isn't there anymore."*

Watch For

Help students map the cause-and-effect chain. Find the feedback loop: declining aquifer reduces food production, but population growth increases demand — a diverging gap. Discuss how climate change compounds the problem by reducing recharge.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three scenarios: live within our means, keep overdrafting, or watch climate change shrink the budget while population grows the bills. Which future feeds 10 billion people?"*

Watch For

Students predict depletion timelines before running simulations. Focus on the current overdraft scenario — identify the year when aquifer levels can no longer support irrigation. Compare Regional Food Production Capacity across scenarios.

Slide 8: Discoveries

5 min

Say: *"Your model just showed why water is the crisis behind the food crisis behind the migration crisis behind the political crisis. It's all connected through this one resource."*

Watch For

Lead discussion on cascading system failures. Connect to real-world examples: Cape Town's Day Zero, California's Central Valley subsidence, India's groundwater crisis. Discuss water as a human right vs. economic commodity.

Slide 9: STEM Challenge

12 min

Say: *"Thirty years from now, this aquifer runs dry. 15% of the nation's food comes from this region. Design the plan that keeps the water flowing — or finds a way to grow food without it."*

Watch For

Groups design comprehensive 30-year water transition plans. Must address efficiency, crop shifts, governance, and economic support for affected farmers. Present as water authority strategic plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Population and Agricultural Demand, Groundwater Extraction Rate
- **INTERNAL:** Aquifer Water Level, Natural Recharge Rate, Surface Water Availability, Regional Food Production Capacity

Relationships

- **Population and Agricultural Demand to Groundwater Extraction Rate:** POSITIVE (+) — Higher population and agricultural demand drives increased Groundwater Extraction Rate as farmers pump more water to irrigate expanding cropland. Agriculture accounts for 70% of freshwater use, and demand grows with both population and per-capita consumption.
- **Groundwater Extraction Rate to Aquifer Water Level:** NEGATIVE (-) — Higher extraction directly depletes Aquifer Water Level by removing water faster than natural processes can replenish it. The relationship is arithmetic: if extraction exceeds recharge, the aquifer declines at a rate equal to the difference.
- **Aquifer Water Level to Regional Food Production Capacity:** POSITIVE (+) — Higher Aquifer Water Level means reliable irrigation supply, supporting full agricultural production. As the aquifer declines, wells must go deeper (increasing cost), eventually reaching depths where pumping is no longer economically viable, and finally running dry entirely.
- **Natural Recharge Rate to Aquifer Water Level:** POSITIVE (+) — Higher Natural Recharge Rate slows or reverses aquifer depletion by adding water back to the system. However, in fossil aquifer systems, recharge is negligible compared to extraction — the natural input cannot compensate for the human output.

Key Discoveries

- Only 0.5% of Earth's total water is accessible freshwater — the rest is saltwater (97.5%) or frozen in ice caps and glaciers (2%) — making freshwater one of the most limited and unevenly distributed resources on the planet
- Major agricultural aquifers including the Ogallala (USA), North China Plain, Upper Ganges (India), and Arabian aquifer systems are being depleted 3-10 times faster than natural recharge, effectively mining fossil water that accumulated over millennia
- Agriculture consumes 70% of all freshwater withdrawals globally, with inefficient flood irrigation wasting up to 60% of applied water — the food system is by far the largest driver of water scarcity
- Water scarcity creates cascading system failures: depleted aquifers reduce food production, food shortages drive migration, migration concentrates populations in cities that face their own water stress — a compounding crisis

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"There's plenty of water — it just needs to be distributed better"	While water distribution is unequal, the fundamental problem in many regions is absolute depletion of groundwater stocks that cannot be replaced on human timescales. Redistribution cannot solve the crisis when total available water is declining. Furthermore, transporting water over long distances is enormously expensive and energy-intensive.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Desalination can solve water scarcity"	Desalination technology is improving but faces fundamental constraints. It requires 3-10 kWh of energy per cubic meter of water, costs $0.50-2.00 per cubic meter (far more than groundwater at $0.05-0.20), produces toxic brine waste, and is practical only for coastal cities. Agriculture — which consumes 70% of freshwater — cannot economically operate on desalinated water. Desalination is part of the solution for drinking water but cannot replace aquifer depletion at scale.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Water is a renewable resource"	Surface water flowing through the water cycle is renewable, but groundwater in fossil aquifers is not. Many of the world's most productive agricultural aquifers were filled during past geological periods and receive negligible recharge today. Pumping this water is extracting a non-renewable resource, no different in principle from mining coal or oil. The model makes this distinction critical: renewable surface water cannot replace non-renewable groundwater at the volumes agriculture demands.